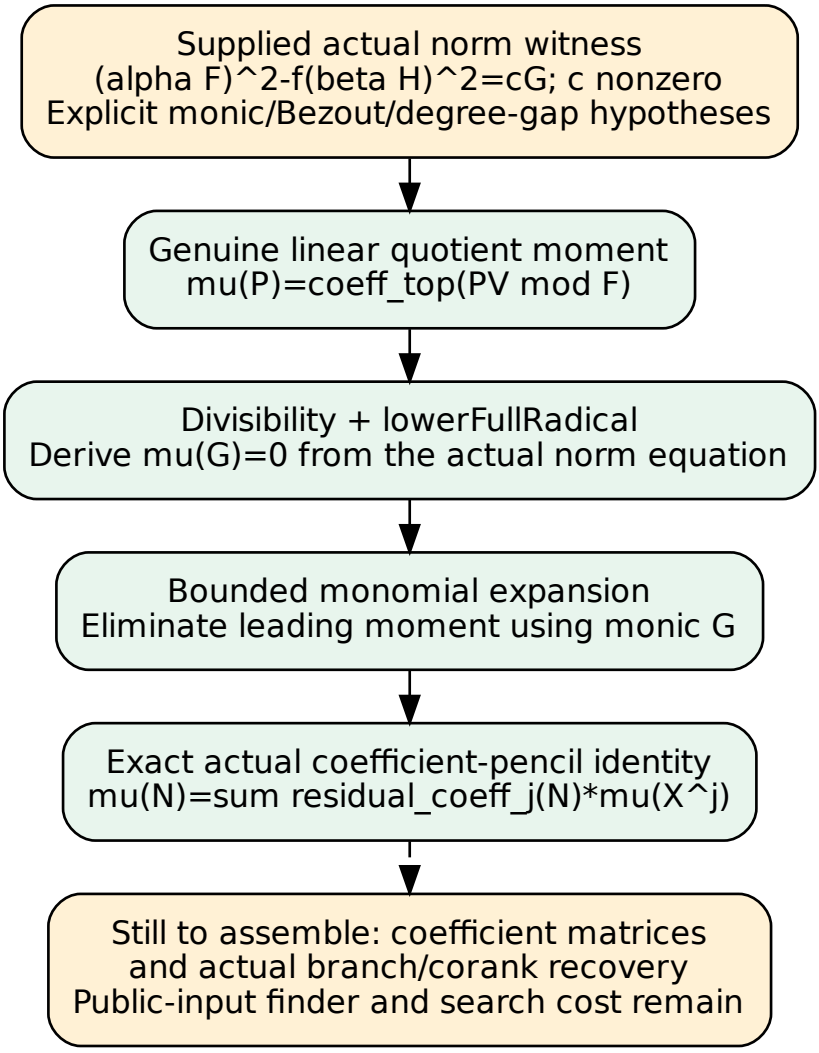
Supplied actual norm witness
(alpha F)^2-f(beta H)^2=cG; c nonzero
Explicit monic/Bezout/degree-gap hypotheses
Genuine linear quotient moment
mu(P)=coeff_top(PV mod F)
Divisibility + lowerFullRadical
Derive mu(G)=0 from the actual norm equation
Bounded monomial expansion
Eliminate leading moment using monic G
Exact actual coefficient-pencil identity
mu(N)=sum residual_coeff_j(N)*mu(X^j)
Still to assemble: coefficient matrices
and actual branch/corank recovery
Public-input finder and search cost remain